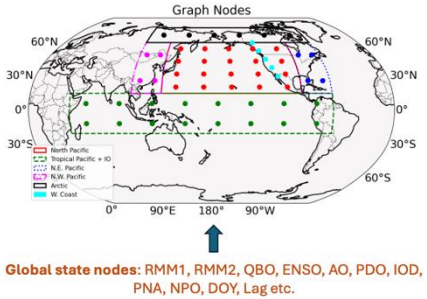
Probing state dependent predictability of W. coast AR activity using a Graph Transformer

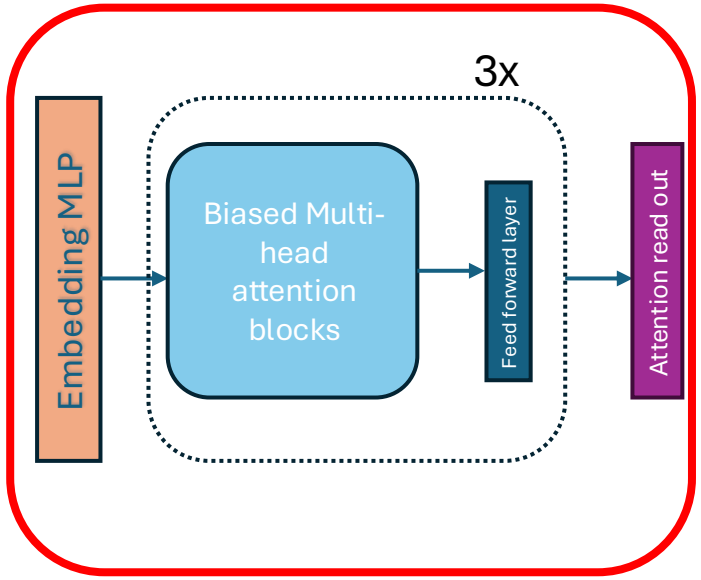
Input Graph



Global state nodes: RMM1, RMM2, QBO, ENSO, AO, PDO, IOD, PNA, NPO, DOY, Lag etc.

Features on mesh + coastal nodes + climate mode indices at t=-i,..0 (in weeks)

Initial attention bias by locality/correlations

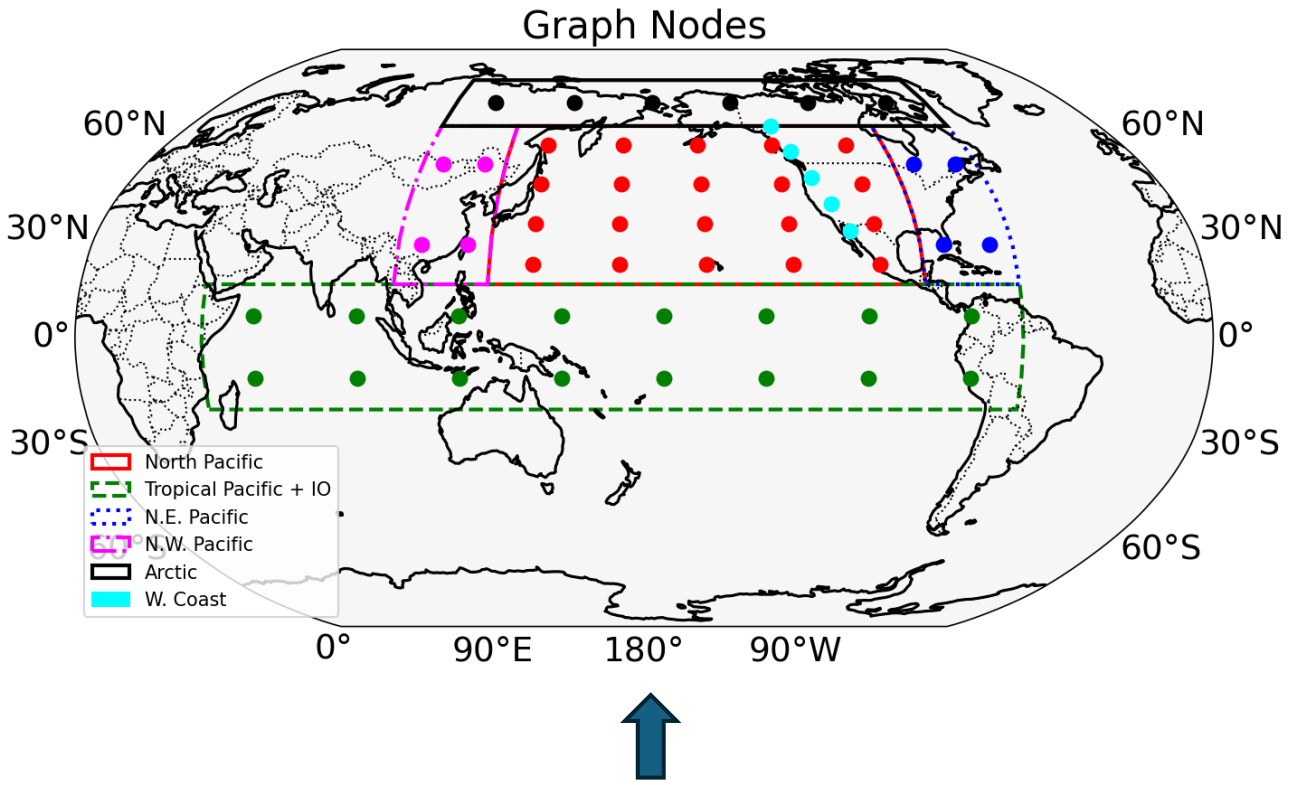


Output (Weeks 1,2,3 and 4)

- *AR frequency anomaly terciles* (above, near, or below-normal) on Coastal nodes
- *Z500 regime in the N. Pacific*
- *Calibrated confidence of predictions*

Basic workflow using a Graph Transformer

Graph Structure



Global state nodes: RMM1, RMM2, QBO, ENSO, AO, PDO, IOD, PNA, NPO, DOY, Lag etc.

Node structure:

- ~60 nodes representing *mesh nodes* relevant to N.Pacific + *coastal nodes* + *climate modes tokens*
- Each mesh/coastal node has features - *regionally and 7-day averaged*
  - *Z500, IVT_x, IVT_y, + tendencies etc.*

Edges:

- Fully connected
  - *Attention-patterns inform strength of node-interaction given different contexts (ex: global states)*

Saved inference on Eval data (Held-out Winters)

- Forecasts
- Confidence
- Attention matrices
- Gradients

Partition on Global States

Ex: MJO x ENSO, MJO x PDO

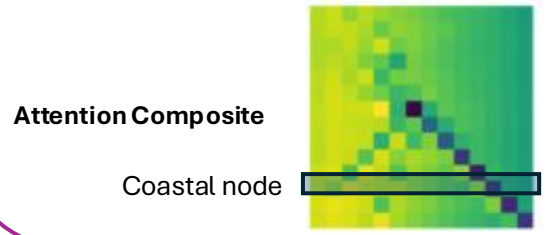
Partition on Confidence levels

Conditional skill and regional variability

Layer-wise Attention Composites

Identifies *sources of information (nodes)* for a *given prediction and lag and/or partition*

Probe significant attention weights along edges from the tropics, ENSO, RMM tokens etc to coastal nodes



Causal attribution (model) using gradients+ ablation studies

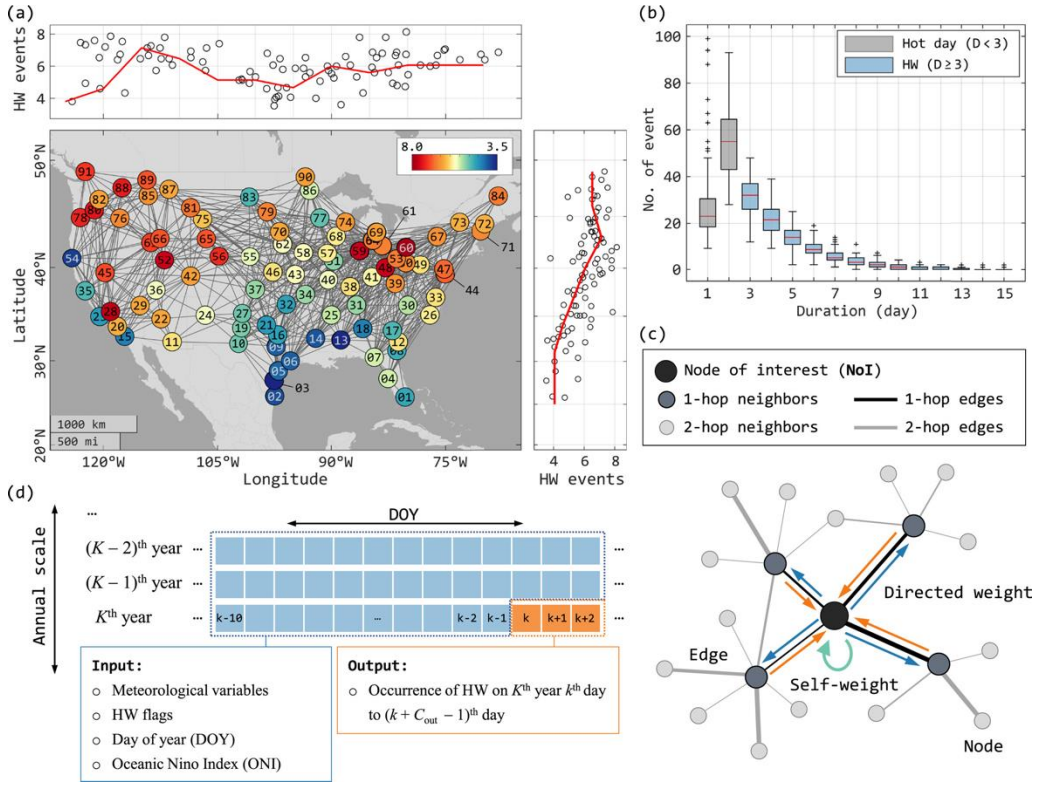
Gradients can probe relative impact of information sources – sensitivity map

Ablation studies – Remove regions/climate modes and retrain to understand loss in skill

Analysis to probe state dependent predictability

Relevant References

- Li, P., Yu, Y., Huang, D., Wang, Z.H. and Sharma, A., 2023. Regional heatwave prediction using graph neural network and weather station data. *Geophysical Research Letters*, *50*(7), p.e2023GL103405.



(a) Geographic locations of the selected 91 weather stations; Filled color in (a) shows the annual average number of heatwave (HW) events at each node. The sub-panels above and to the right of (a) show the number of HW events with longitude and latitude, respectively, with red solid lines showing the moving average. (b) Statistics of HW frequency and duration over all stations from 2006 to 2020 (15 years). (c) A diagram illustrating network structure centered on the node of interest; The value and direction of the attention weights on edges in (c) are represented by the thickness of the lines and arrows. (d) A diagram illustrating the model input and output in temporal scale when C_{in} = 10, C_{out} = 3, N_{hist} = 2.

Relevant References

- Prapas, I., Papadopoulos, N., Bountos, N.I., Michail, D., Camps-Valls, G. and Papoutsis, I., 2026. TeleViT1. 0: teleconnection-informed vision transformers for subseasonal to seasonal wildfire pattern forecasts. *Machine Learning: Earth*, *2*(1), p.015019.

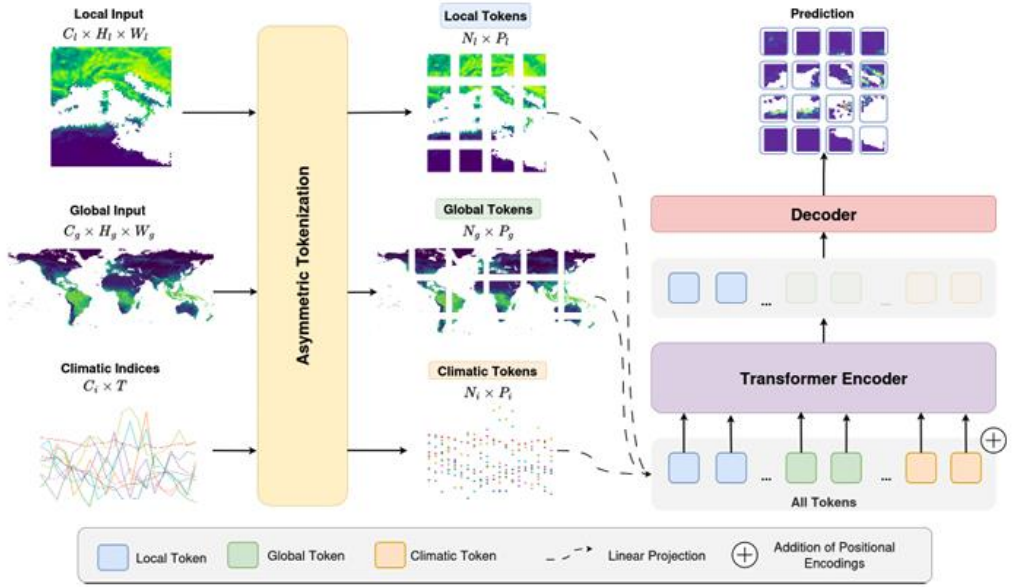


Figure 1: Architecture of the TeleViT model. The model processes three types of inputs (local, global, and indices) through an asymmetric tokenization strategy, followed by a transformer encoder and linear decoder to produce predictions.

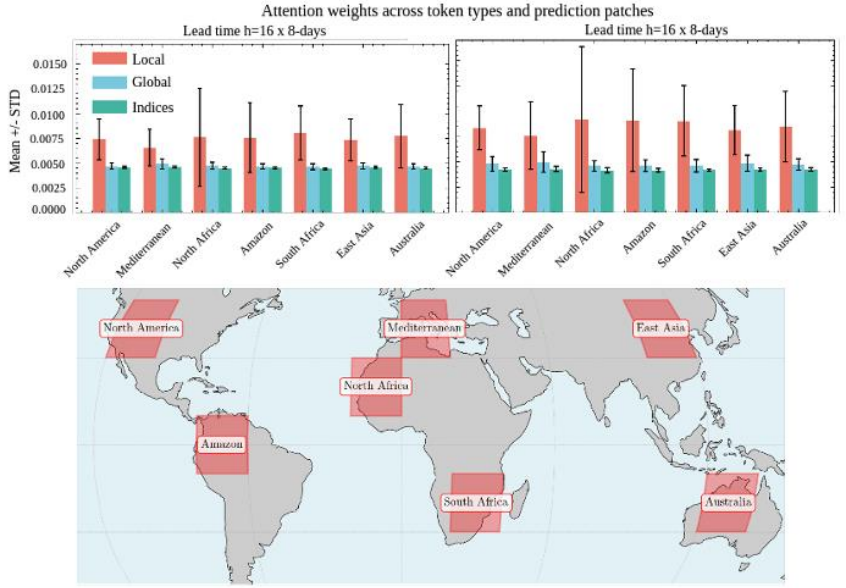


Figure 5: Bar charts of mean attention weights with standard deviations (error bars) for each token type (local, global, indices) across prediction patches around the world. The left chart shows results for a forecasting horizon of $h = 0 \times 8$-days, while the right chart shows results for $h = 16 \times 8$-days. The bottom map indicates the geographical locations of the selected patches.

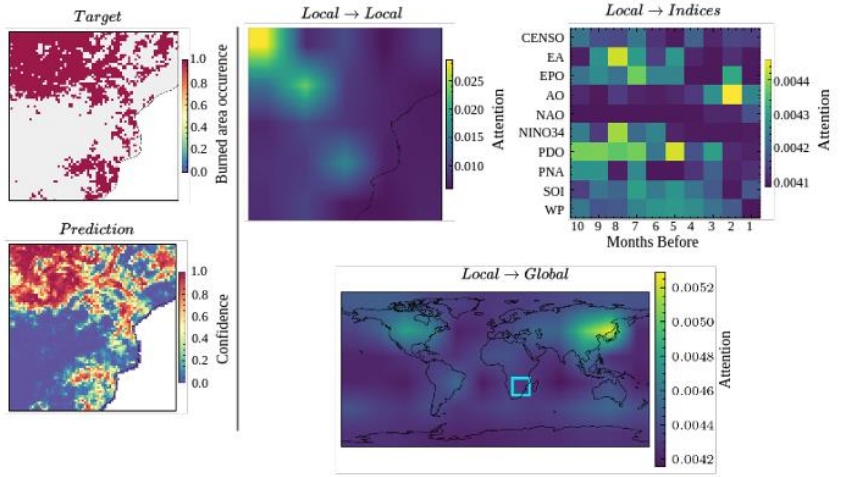


Figure 6: Left side: Prediction vs target (first row), local-to-local attention (middle left), local-to-indices attention (middle right), local-to-global attention (bottom)